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DEPARTMENT OF
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Subject : Fundamentals of Telecommunication Systems

Experiment Number : 01

Experiment Title : Study and Operation of a Basic Telephone System

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Experiment No. 01

Title: Study and Operation of a Basic Telephone System

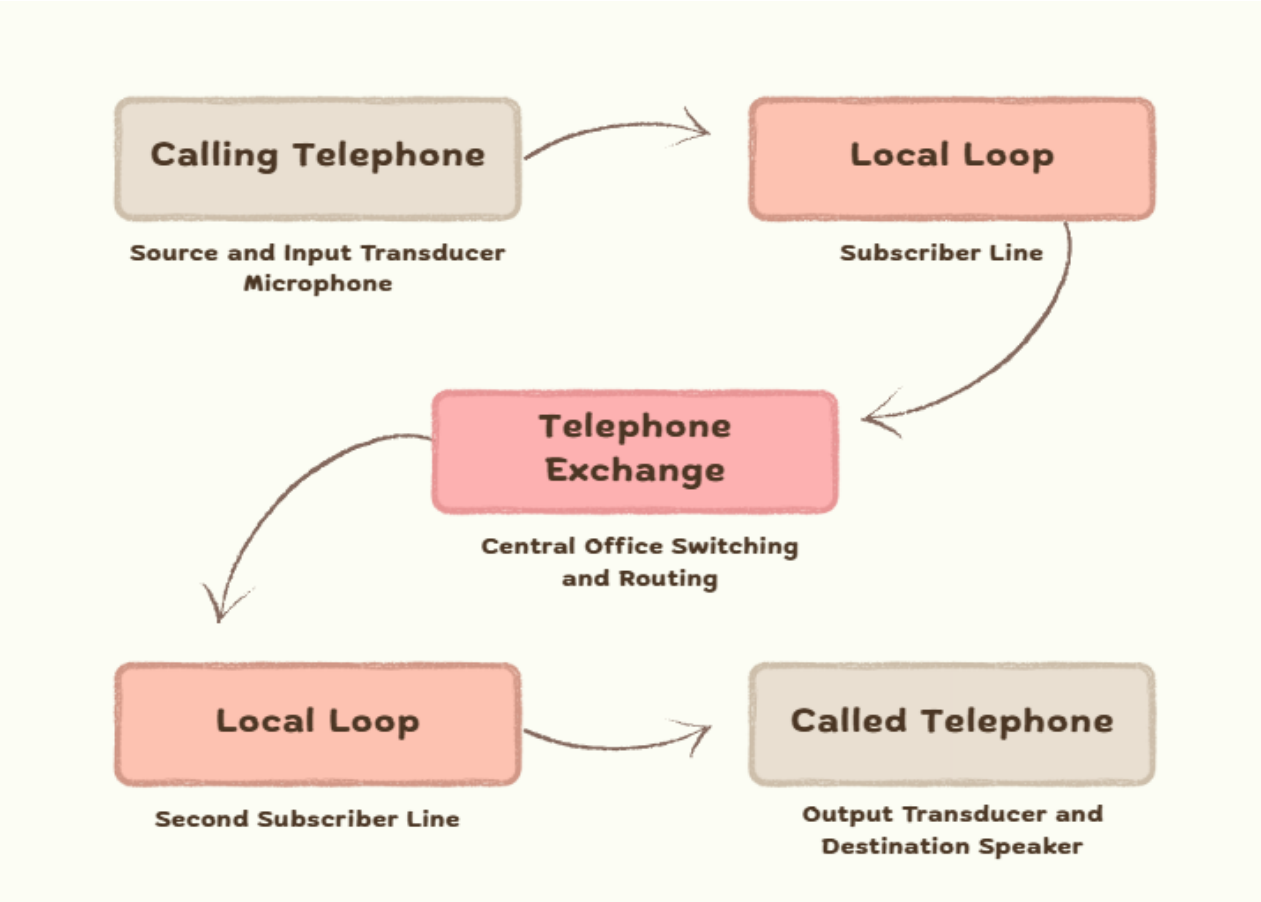
Aim:

To study and understand the construction, components, and operation of a basic telephone system using online resources and videos.

Theory

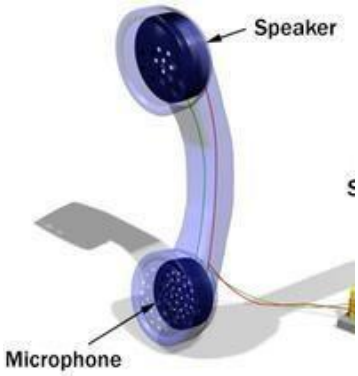
A basic telephone system enables voice communication between two distant users by converting speech signals into electrical signals, transmitting them through a communication network, and reconverting them back into sound at the receiver.
Traditional telephony operates on the Public Switched Telephone Network (PSTN), which uses circuit switching to establish a dedicated communication path for the duration of a call.

Block Diagram

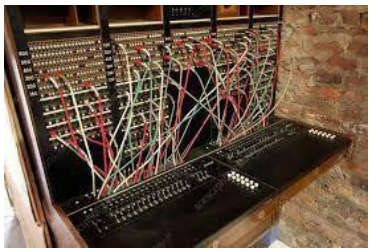


Basic Mobile Communication System

Task 1: The "Inside the Handset" Investigation

Component	Visual Reference	Technical Function
Microphone		The microphone in a telephone handset converts the user's voice into electrical signals. These signals are sent through the telephone line to the telephone exchange and then to the receiving handset.
Speaker		The speaker in a telephone handset converts incoming electrical signals into sound waves. This allows the user to hear the voice of the person calling.

Switch Hook		The switch hook in a telephone handset detects whether the handset is lifted or placed on the cradle. Lifting the handset signals the exchange to start a call, while placing it back ends the call.
Ringer		The ringer in a telephone handset produces a ringing sound when an incoming call is received. It is activated by a ringing signal sent from the telephone exchange.
Keypad		The keypad on a telephone handset allows the user to dial numbers and enter commands. Each key press sends a corresponding signal to the telephone exchange for call processing.

Exchange		The telephone exchange connects one telephone handset to another by switching communication paths. It handles call setup, routing, and disconnection between subscribers.

Task 2: Basic Blocks of a Telephone System

1. Microphone (Transmitter):
   The microphone in a telephone handset converts the user's voice into electrical signals. These signals vary according to the sound pressure of the voice. This conversion is essential for transmitting speech over the telephone network.
2. Local Loop:
   The local loop is the physical wired connection between the telephone handset and the telephone exchange. It carries voice signals, dialing information, and control signals. The local loop plays a crucial role in providing power and communication to the handset.
3. Telephone Exchange:
   The telephone exchange is a central switching system that connects calls between subscribers. It interprets dialed numbers and establishes a communication path. The exchange manages call setup, supervision, and termination.
4. Transmission Network:
   The transmission network links multiple telephone exchanges over long distances. It carries voice and signaling data using cables, optical fiber, or wireless links. This network enables communication between subscribers located in different regions.
5. Receiver (Speaker):
   The receiver in a telephone handset converts incoming electrical signals into audible sound. It reproduces the caller's voice clearly for the user. This block completes the communication loop at the receiving end.

6. Ringer and Dialing Circuit:
   The ringer circuit alerts the user of an incoming call by producing a ringing sound. The dialing circuit generates dialing signals (pulse or DTMF tones) when numbers are pressed. Together, they support call notification and call initiation.

Task 3: Conceptual Working of a Telephone System Off-Hook Condition:

1. Reception of Dial Tone:
   After detecting the off-hook condition, the telephone exchange sends a dial tone to the handset. The dial tone informs the user that the system is ready to accept dialing input. This confirms a successful connection to the exchange.
2. Dialing (DTMF Generation):
   When the user presses keys on the keypad, DTMF tones are generated. These tones are transmitted to the exchange through the local loop. The exchange decodes them to identify the dialed number.
3. Switching & Routing:
   When the user presses keys on the keypad, DTMF tones are generated. These tones are transmitted to the exchange through the local loop. The exchange decodes them to identify the dialed number.
4. Ringing State (Ring-back vs. Ringing):
   The calling user hears a ring-back tone indicating the call is being processed. At the same time, the called telephone receives a ringing signal. This continues until the call is answered or terminated.
5. Voice Transmission (The Talking State):
   Once the call is answered, two-way voice communication begins. The microphone and speaker in both handsets continuously convert voice signals. The exchange maintains the connection throughout the conversation.
6. On-Hook (Call Termination):
   When either user places the handset back, the telephone enters the on-hook condition. The exchange detects this and disconnects the call. All network resources used for the call are released.

Conclusion

This experiment helped in understanding the complete working of a telephone system from the handset to the network. The microphone in the telephone handset converts the user's voice into electrical signals, which are transmitted through the local loop to the telephone exchange. The exchange processes the dialed information, performs switching and routing, and sends the signals through the transmission network to the called subscriber. At the receiving end, the speaker reconverts the electrical signals back into sound, enabling two-way voice communication. Thus, the telephone system efficiently demonstrates how physical voice signals are converted, transmitted, and reconstructed using electrical and network principles.